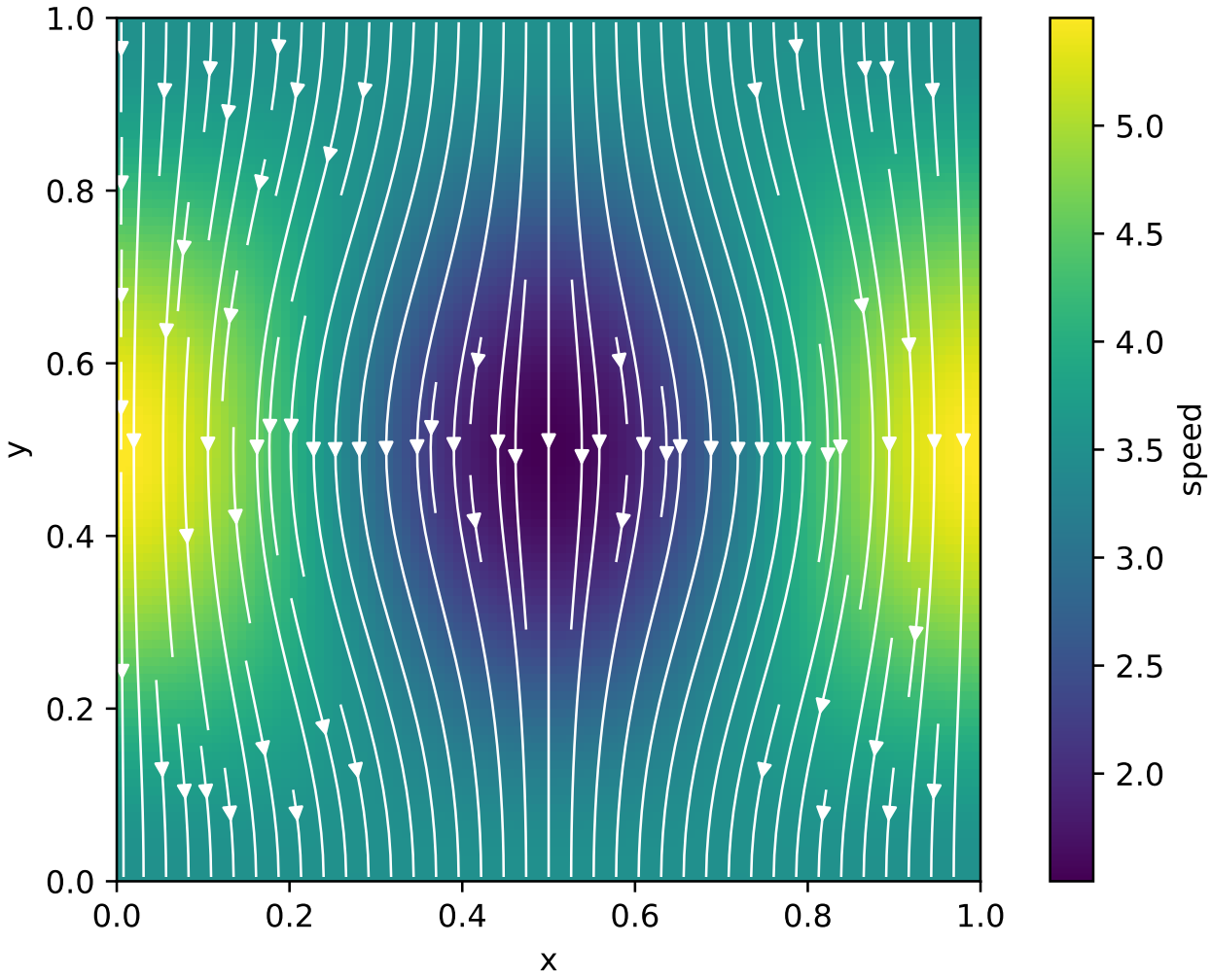
Sean Gravelle
Numerical PDEs Homework #1

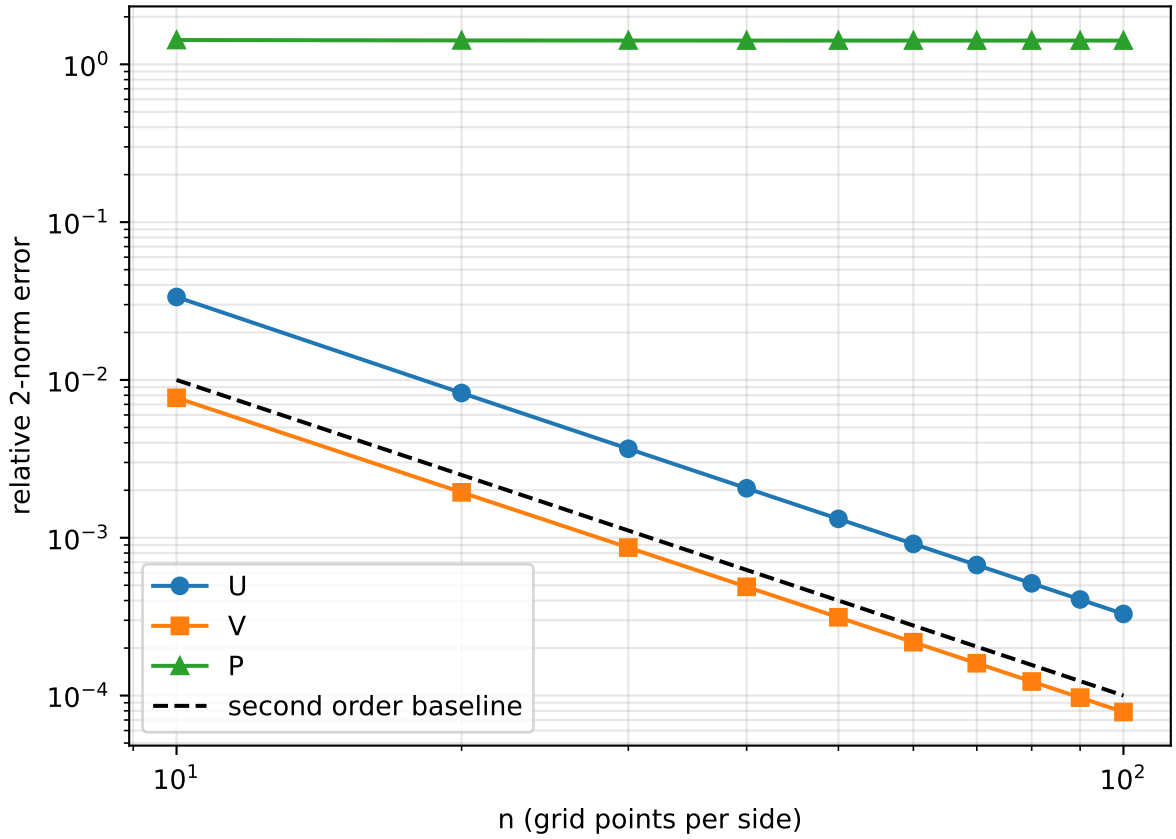
Problem 1: Solve Stokes

My solution to the Stokes equations in 2 dimensions, with the given boundary conditions, is plotted below. I used $n = 100 = Nx = Ny$ gridpoints in both x and y directions. Arrows indicate direction of the fluid's flow and color indicates the magnitude of the solution vector; that is, the speed of the fluid.



Problem 2: Error Analysis

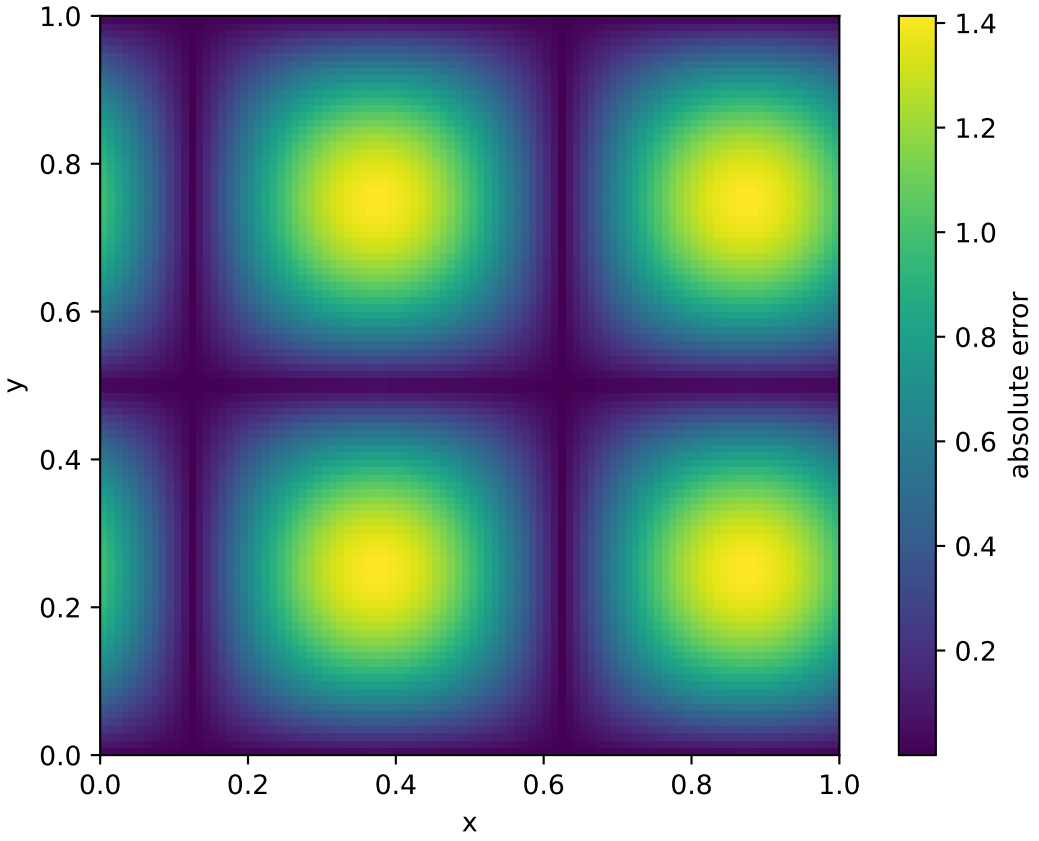
Plotting the relative error of numerical solutions on an $n \times n$ grid gives the following errors for U, V, and P. The slope of the errors for U and V shows that their errors decrease proportionally to $1/n^2$. The error for P, in contrast, is much higher and constant in n, which is unexpected.



Problem 2: Error Analysis, continued

Identifying source of error in P

Since the error in P identified in the figure above is both large and constant in P, this suggests it may come from a constant difference between the analytical and numerical solutions for P. However, this is not the case, because I subtracted the mean value of P from my numerical solution. I also verified this by plotting the absolute error of P as a function of (x, y), holding $n = 100$ fixed. This plot is shown below; here, I measured error simply by taking the absolute value of the difference between the numerical and analytical solutions of P. This plot shows that the error is not at all constant in space, so the source of the high error in my solution to P comes from something other than a constant difference.



AI disclosure statement: All plots were made with LLM assistance. All mathematical work and all code to produce the numerical solutions and error calculations are my own, without LLM.